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Original Research

Frontal plane pelvic kinematics during high velocity running: Association with hamstring injury history

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ABSTRACT

Introduction: Hamstring injuries are the most prevalent non-contact soft tissue injury in sports, with a larger portion of injuries being recurrent. The sagittal plane running kinematics correlated to hamstring injury history has been well documented. However, analysis of frontal plane kinematics allows for observation of stability and symmetry. This study aimed to examine the frontal plane running kinematics of elite collegiate level sprinters, with and without previous hamstring injury, compared to healthy counterparts.

Methods: Thirty-nine participants performed three 50-m sprints, with three inertial measurement unit sensors placed on the pelvis: one on each iliac crest and one on the sacrum. Participants were classified based on sex, competitive status, and injury history. To investigate differences based on group classification, the data were used to analyze mediolateral motion (relative magnitude of mediolateral acceleration) and asymmetry (difference in acceleration between right and left iliac crests) during each stance phase throughout the run.

Results: Injured sprinters displayed significantly greater mediolateral motion and asymmetry during stances than healthy counterparts.

Conclusions: This study demonstrates that frontal plane running stance dynamics are different in athletes with previous hamstring injury than healthy athletes. These athletes may benefit from rehabilitation strategies targeting postural control and stability during dynamic tasks.

1. Introduction

Hamstring strain injuries are the most prevalent non-contact injury in Australian football (Orchard & Seward, 2002), American football (Feeley et al., 2008), rugby union, (Brooks et al., 2006), soccer (Ekstrand et al., 2011; Woods et al., 2004) and sprinting (Bennell & Crossley, 1996; Drezner et al., 2005). In the NCAA alone, 1142 hamstring strains were reported from 2010 to 2014 (Dalton et al., 2015a) and athletes who sustain a hamstring strain are more likely to re-injure when they return to sport (Devlin, 2000; Sherry & Best, 2004; Sherry et al., 2011, 2015). The high rate of hamstring strain re-injury points to possible underlying differences between athletes with previous injury and their healthy counterparts. Altered kinematics of injured runners such as increased anterior pelvic tilt (Hennessy & Watson, 1993) and knee extension during terminal swing (Chumanov et al., 2007) have been cited as a potential influence on the high rates of re-injury.

Running kinematic studies have helped to inform potential injury events in stance and swing phase, but primarily analyze sagittal plane movements. From these studies it has been determined that, during running the hamstrings are most susceptible to injury during the terminal swing and initial contact phases (Kenneally-Dabrowski et al., 2019; Orchard, 2002a; Verrall et al., 2003; Yu et al., 2008). In terminal swing, the hamstrings are required to eccentrically contract to decelerate knee extension and hip flexion (E.S. et al., 2007a; Orchard, 2002b) while also reaching their maximum stretched length during the terminal swing phase of running gait (E.S. et al., 2007b). However, during initial contact, the hamstring muscles generate large hip extension and knee flexion torques (Mann, 1981). The large ground reaction forces and joint torques present at initial contact, coupled with the lengthening of the hamstring during the preceding terminal swing phase may play a critical role in potential hamstring injury mechanisms. Analysis of frontal plane kinematics during stance can aid in the examination of symmetry and

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balance (Houck et al., 2006; Williams et al., 2016), helping to illuminate the potential injury mechanisms that arise during initial stance.

Pelvic drop is an example of a frontal plane stance event that can increase stress in the thigh muscles (Winter et al., 2019a). Pelvic drop involves a lowering of the contralateral side of the pelvis during single leg stance (i.e. the pelvis is lower on the side in swing phase than the side in stance). This causes the stance knee to move into a valgus position (Kozinc & Šarabon, 2017a; Powers, 2010a), a position that increases risk of several knee pathologies, including anterior cruciate ligament ruptures and patellofemoral pain (Herrington, 2014; C. F. Lin et al., 2012). The hamstrings and quadriceps are activated to stabilize the knee in valgus and varus positions (Hirokawa et al., 1991; Lloyd et al., 2005) and thus the force requirements of the hamstrings to stabilize the valgus knee position due to pelvic drop could increase tensile stress in the muscle during ground contact. Additionally, pelvic drop causes decreased stability and is often accompanied by a lateral trunk lean which is used to regain balance (Kozinc & Šarabon, 2017b; Novacheck, 1998). This lateral trunk lean could move the center of mass outside the base of support during single leg stance, requiring a counterbalance torque from their gluteal and thigh muscles to stabilize themselves. Due to the increased internal and impact forces present during sprinting, the potential extra requirements of the hamstrings in those who exhibit pelvic drop may be a possible cause for repeated muscle injury.

Running kinematics have been shown to be affected by sex differences and females are reported to be twice as likely to sustain running injuries than male counterparts (Taunton et al., 2002). Females demonstrate greater hip adduction and internal rotation (Ferber et al., 2003) and exhibit greater knee valgus angles (Malinzak et al., 2001) than males during long distance running. Pelvic drop during stance could contribute to increased frontal plane pelvis position and knee valgus motion in females when running long distances. If differences in body composition or joint laxity (Lewis et al., 2017) between males and females are the root cause of the differing pelvis kinematics when running, these differences will persist even as speed increases. However, increased hip adduction and internal rotation has been linked to weak lower limb muscle strength (Ireland et al., 2003; Leetun et al., 2004; Niemuth et al., 2005; Rodrigues et al., 2020), therefore, examining the differences in elite level athletes could reduce the differences in these metrics. Additionally, it has been demonstrated that the stiffness of the lower limb increases with speed (Arampatzis et al., 1999), this increase of stiffness could result in less pelvis movement in general, thus reducing the differences between the sexes.

The objective of our work was to investigate how frontal pelvic kinematic metrics related to pelvic drop differed based on sex, competitive status, and injury history. We hypothesized that 1) unlike distance running, men and women display similar pelvic kinematics when sprinting and 2) elite sprinters with hamstring injury history display asymmetry in pelvic kinematics compared to healthy counterparts.

2. Methods

2.1. Participants

Thirty-nine participants ages 18–30 years old were recruited for this study: 18 females, 21 males (self-reported sexes). Participants were categorized into sprinter (10 female, 12 male) and control (8 female, 9 male) populations. Sprinter participants were recruited specifically from NCAA Division 1 track and field teams. Control participants were defined as participants who were not involved with NCAA Division 1 athletics. Participants were asked to fill out a survey to assess their injury history, particularly sites and frequency; participants who had experienced hamstring injury within the last 2 seasons were categorized within the Sprinter (7 previously injured male, 5 previously injured female) and Control (1 previously injured male, 2 previously injured female) specifications. Exclusion criteria for the study included any medical condition or current injury that may affect running. Participants provided written informed consent before participating in the study, and assent was obtained from the coaching staff at NCAA institutions. IRB approval was obtained from the governing institution.

2.2. Protocol

Data collection was performed at university indoor track and field facilities and consisted of three 50 m sprints with a walk back recovery. The distance was chosen as it allows for maximum velocity to be achieved but is not long enough that the results would be confounded by fatigue while running. Between sprints, participants were instructed to take as much rest as required to ensure that they were completing the subsequent runs at the same level as the first. Participants wore their own clothing and shoes. Control participants were given a warm-up to complete to ensure they were prepared for the study requirements. The warm-up consisted of dynamic stretches, running drills and jogging. Sprinter participants took part in the study after completing their team's usual warm up routine.

2.3. Data collection

The motivation of this study was to analyze pelvis motion elite sprinters in real-world track and field environments, as such we used IMU sensors placed around the lumbo-pelvic complex, with one sensor taped onto each iliac crest and one sensor taped onto the sacrum (Fig. 1). A wireless inertial measurement unit (IMU) system (Opal wireless sensors, APDM Inc., Portland OR, USA) was used to collect the motion data from the trials. The sensor system consisted of three Opal sensors (dimensions: 36.5 mm 36.1 mm 13.4 mm), each equipped with a tri-axial accelerometer, gyroscope, and magnetometer, and configured for wireless synchronized logging at a sampling rate of 128 Hz. Prior to data collection, calibration was performed for each sensor as per the instruction manual.

2.4. Data preparation

Since each sensor has its own independent coordinate system, data from different sensors cannot be related without first being transformed to a shared coordinate system (Cole et al., 1993; Horenstein et al., 2019a). In order to perform comparative analysis of the acceleration, the local independent coordinate system of each iliac crest sensor was mapped to the global coordinate system using the ADPM calculated quaternions and then from the global coordinate system to the sacrum sensor coordinate system (Fig. 1). Rotation matrices calculated using the quaternion output of the sensor data recording were used to map the coordinate system of the data using methods described in Horenstein et al. (Horenstein et al., 2019b). Analysis of acceleration data was performed using a custom written program in MATLAB R2018a (The MathWorks Inc., Natick, MA).

The data were segmented into individual steps using the pattern of local minima and maxima of vertical acceleration during a running stride using methods described in Winter et al. (Winter et al., 2019b). The segmented strides were normalized to percent stance with 0 and 1 representing initial contact and toe-off respectively. The 15 steps identified at the beginning and end of the trial were excluded from analysis to account for variabilities that could be introduced due to acceleration and deceleration phases. For each segmented stride we calculated metrics that could underlie pelvic drop, increases in mediolateral motion

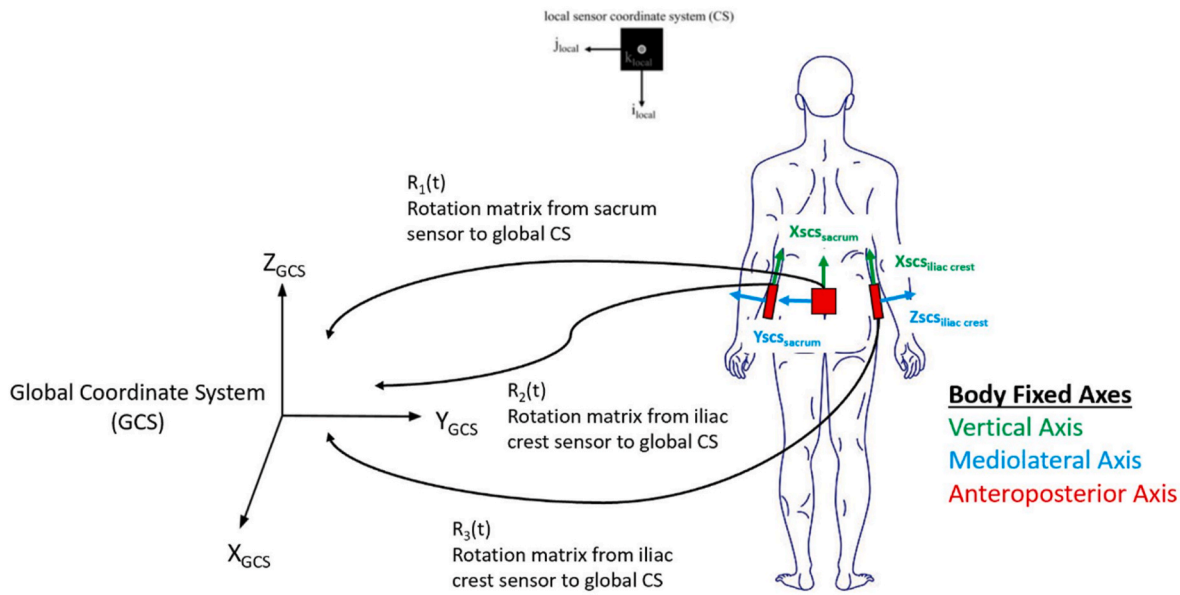

Fig. 1. IMUs were used to create a sensors-based motion capture system. The IMUs were placed on the sacrum and on each iliac crest and were secured with elastic athletic tape. The conjugate of the normalized quaternions was used to rotate the sensors' coordinate systems to the global coordinate system.

and asymmetrical acceleration of the pelvis.

2.5. Mediolateral motion

Mediolateral motion in this study was defined as the portion of mediolateral acceleration of the sacrum sensor during single leg stance of running. Pelvic drop is accompanied by a lateral trunk lean (Kozinc & Šarabon, 2017b; Novacheck, 1998) which can cause a decrease of stability by moving the center of mass outside the base of support during single leg stance. Excessive movement in the mediolateral direction can be an indicator of greater lateral trunk lean, dynamic instability and less efficient running movement (S. P. Lin et al., 2014).

Mediolateral motion was measured using the root mean square ratio (RMSR); RMSR has been used to measure dynamic postural stability (Winter et al., 2019b) and gait abnormality (Sekine et al., 2013) during locomotion. To calculate the RMSR we used methods described in Sekine et al., we first calculated the root mean square total (RMST), which is the resultant of the root mean square (RMS) acceleration in each direction.

$$\mathrm{RMST} = \mathrm{sqrt}(\mathrm{RMSx}\hat{2} + \mathrm{RMSy}\hat{2} + \mathrm{RMSz}\hat{2})$$

RMSx, RMSy and RMSz are the root mean square of sacrum sensor acceleration in the x, y and z directions. We then calculated the ratio in the x-direction (medial/lateral) for each ground contact to indicate the contribution of mediolateral acceleration to overall acceleration.

$$\mathrm{RMSRx} = \frac{\mathrm{RMSx}}{\mathrm{RMST}}$$

2.6. Pelvic asymmetry

Pelvic drop is observed when one side of the pelvis lowers during single leg stance, meaning that side of the pelvis travels a greater distance from the neutral position. Therefore, we can identify pelvic drop

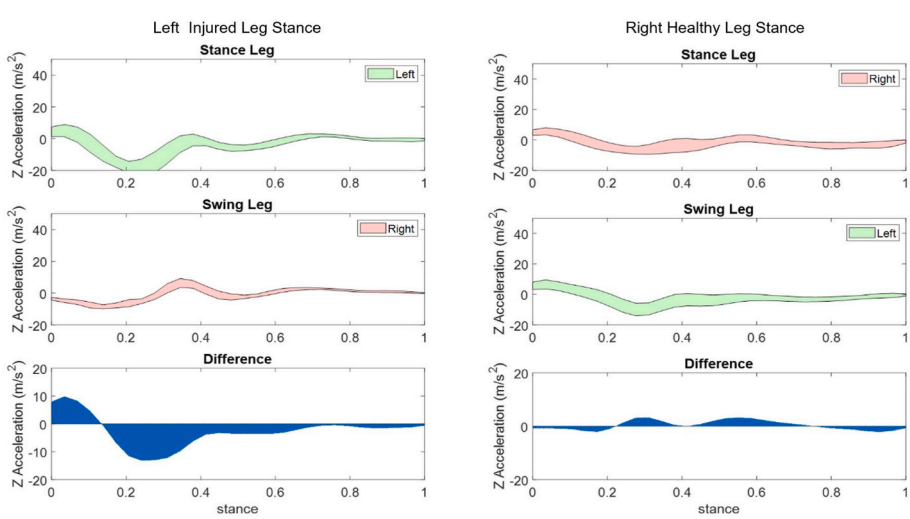

Fig. 2. Example of pelvic asymmetry measure for a single participant. Pelvis asymmetry defined as difference between stance and swing iliac crest sensor acceleration. Top panels depict the mean and 95% confidence interval, and the bottom panels depict the pelvic asymmetry metric. When the right (healthy) leg was on the ground, the vertical (z) acceleration of the pelvis is the same on each side. When the left (injured) leg was on the ground there was a large difference in the vertical acceleration between right and left iliac crest sensors indicating pelvic asymmetry.

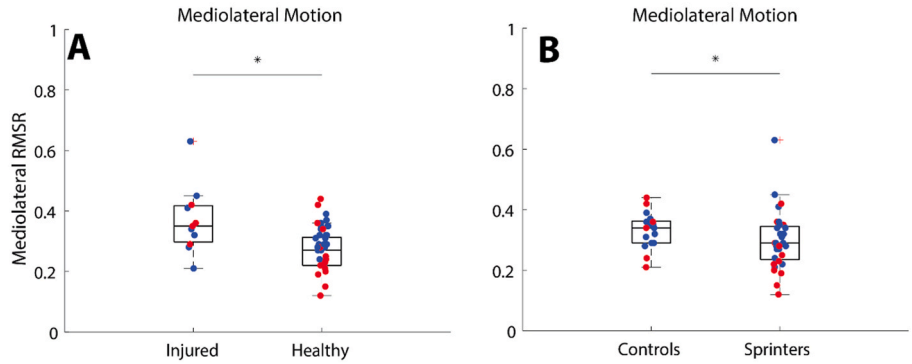

Fig. 3. Mediolateral motion results. (A) Previously injured sprinters exhibited greater mediolateral motion during stance phase than healthy sprinter. (B) Controls displayed greater mediolateral motion during stance than Sprinters. The box indicates the standard deviation, the line indicates the median and significant differences are denotes with an asterisk (*). The red points indicated female participants and the blue points indicate male participants. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

by observing increased asymmetry of frontal plane motion during stance. *Pelvic asymmetry* was defined as asymmetry in the vertical acceleration between right and left iliac crest sensors during single leg stance.

To calculate *pelvic asymmetry* the mean vertical waveforms of the ipsilateral (stance) and contralateral (swing) sides of the pelvis were obtained for each stride during right and left stance. Through subtracting the swing leg iliac crest vertical acceleration data from the stance leg iliac crest vertical acceleration data, a difference waveform was derived. The absolute area under the difference waveform discerned whether asymmetry was present (Fig. 2). If a participant were completely symmetric, the difference between vertical accelerations of right and left legs during single leg stance would be negligible, resulting in an asymmetry metric of zero (Fig. 2B).

2.7. Statistical analysis

Differences between groups were assessed using a three-way repeated measures Analysis of Variance (ANOVA), (sex: male vs female, competitive status: controls vs sprinters, injury: previously injured vs healthy) for each dependent variable measured. Due to the small sample size, effect size, using Cohen's d, was also implemented for the interpretation of the results. Effect sizes of less than 0.5 are interpreted as small, 0.5–0.8 are considered medium and larger than 0.8 is large (Cohen, 1995).

3. Results

3.1. Mediolateral motion

Mediolateral motion was higher during stance of previously injured sprinters than that of healthy sprinters ($p = 0.02$, $d = 0.93$) (Fig. 3, Table 1). For both sexes, greater mediolateral motion was observed in controls than sprinters ($p = 0.04$, $d = 0.41$). There was no significant difference found between the sexes.

3.2. Pelvic asymmetry

Pelvic asymmetry was higher during stance on the previously injured sprinters than stance on the healthy sprinters ($p < 0.001$, $d = 1.39$) (Fig. 4, Table 1). Controls displayed larger pelvic asymmetry than sprinters during single leg stance ($p < 0.001$, $d = 0.25$). The sprinter population, however, had many outliers most of whom were previously injured.

4. Discussion

This study used acceleration measures collected during track sprinting to investigate the differences in pelvis motion between groups of people of different sexes, injury histories and athletic skill levels. The results of this study provide insight into possible trends in frontal plane sprinter kinematics that could provide insight into injury and re-injury mechanisms.

During the single leg stance phase of sprinting, participants with previous hamstring injury exhibit greater mediolateral motion than healthy participants. It has been shown that individuals who exhibit a large mediolateral RMSR during locomotion have an increased dynamic postural instability due to lumbo-pelvic sway in the mediolateral direction (Le Bris et al., 2006). Since running primarily occurs in the sagittal plane, an increase in postural movement in the mediolateral direction suggests an increase in energy usage not contributing to forward motion (Foch & Milner, 2014). Meardon et al., in 2016 investigated the influence of injury on dynamic postural control in runners (Meardon et al., 2016). They found that compared to healthy runners, injured runners demonstrated less dynamic control of vertical forces when performing various single leg and stabilization tasks. Due to the retrospective nature of both Meardon's study and our study, however, it is still unknown whether this difference in dynamic control caused the injury or is a result of the injury. Nonetheless, we do know that there is a correlation between dynamic postural control and injury history.

Sprinters with previous injury displayed greater asymmetry than healthy sprinters. Asymmetrical acceleration of the sides of the pelvis

Table 1
Measurement results for sex, injury, and competitive subgroups.

Measurement	Mean ± SD						p values		
	Male InjuredSprinters	Male HealthySprinters	Male Controls	Female InjuredSprinters	Female HealthySprinters	Female Controls	Sex	Injury	Competitive Status
Mediolateral Motion	0.38± 0.14	0.30± 0.04	0.33± 0.04	0.36±0.05	0.27± 0.08	0.34± 0.12	0.42	**0.02**	**0.04**
Asymmetry	3.43±1.79	1.34±0.45	2.86± 0.82	2.68±1.96	1.21±0.55	2.01±1.07	0.08	**< 0.001**	**<0.001**

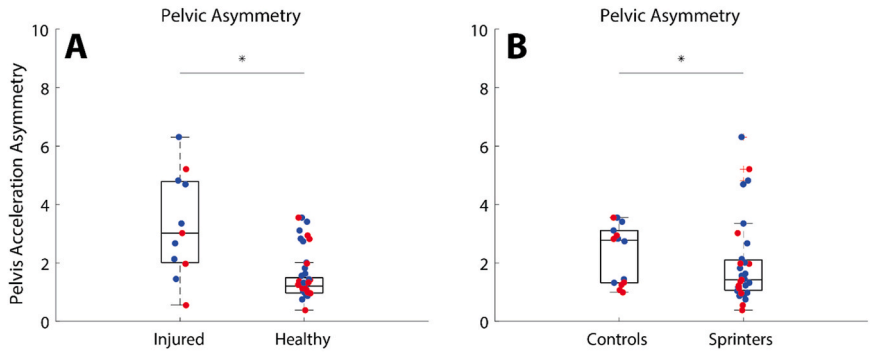

Fig. 4. Pelvic Asymmetry results. (A) Previously injured sprinters exhibited greater asymmetry during stance phase than healthy sprinters. (B) Controls displayed larger asymmetry during stance than Sprinters. The box indicates the standard deviation, the line indicates the median and significant differences are denotes with an asterisk (*). The red points indicated female participants and the blue points indicate male participants. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

was the definition of pelvic drop in this study, meaning sprinters with previous injury exhibited pelvic drop in the frontal plane. *Pelvic drop* has been postulated to arise from motor control adaptation (Noehren et al., 2011) and/or a decreased hip abduction strength (Powers, 2010b). This could point to a decrease in dynamic motor control and functional hip strength during stance phase of the athletes who sustain hamstring injuries, further supported by our findings of greater mediolateral motion during injured leg stance. During single leg stance, stabilizing muscles (i.e., the gluteus medius) are recruited to help with balance (Chumanov et al., 2008). Females activitate these stabilizing muscles to a greater extent than males during walking and jogging (Chumanov et al., 2008). Increasing activation of the stabilizing muscles as speed increases could lead to greater ability to mitigate excessive pelvis motion in the frontal plane. This difference in gluteus medius activation found between the sexes is a possible source of the increased pelvic drop for injured runners when compared to healthy counterparts. There could be a decrease of gluteus medius (or other stabilizing musculature) recruitment during single leg stance, leading to large deviations in frontal plane pelvis motion. To our knowledge this study is unique in its investigation of frontal plane pelvic kinematics during sprinting. The elucidation of the possible relationship between pelvic drop and hamstring injuries could advance our understanding of the movement and rehabilitation of high-level athletes.

Acute hamstring strains typically result in significant rehabilitation time but have an increased risk of re-injury (Sherry & Best, 2004; Sherry et al., 2011), with approximately one third of hamstring strains being recurrent injuries (Dalton et al., 2015b; Sherry & Best, 2004; Sherry et al., 2015). This study identified that sprinters with previous injury exhibit different running stance leg mechanics than healthy sprinters. Though we do not have information on the rehabilitation protocols undergone by the previously injured sprinters, we do have the data to support that they exhibit strength or coordination deficits in their lumbo-pelvic musculature through our asymmetry and stability metrics. It has been proposed that addressing deficits in the pelvic and trunk musculature could decrease the potential of hamstring reinjury (Cameron et al., 2009; Panayi, 2010; Sherry & Best, 2004). With the use of IMU sensors, running kinematics can be measured in real world environments, allowing for on-the-spot assessment of movement abnormalities. Assessing side-to-side differences in lumbo-pelvic strength and coordination could be used to further confirm an athlete's readiness to return to sport. The methods presented here can be refined into a practical and affordable system for physical therapists and athletic trainers to evaluate the functional capabilities of their patients lumbo-pelvic complex during running.

4.1. Limitations

In this study we examined differences between sprinters with previous injury, healthy sprinters, and non-competitive controls. Though we found differences in pelvic asymmetry and mediolateral motion between previously injured and healthy individuals, due to the nature of the study we are unable to ascertain whether these differences contributed to the injury or resulted from the injury. To evaluate this, a longitudinal study is required to assess how athletes run before and after injury. In an ideal study, an athletic team would be followed for a complete season. Using the IMU placements in this study coupled with EMG measuring activation of the hamstrings, quadriceps, and gluteus medius muscles of both legs, comprehensive data of running kinematics and activation could be obtained from each athlete on a regular basis.

The sensors we used in the study had a sampling rate of 128Hz, since we tested elite sprinters as well as controls in our dataset there is a possibility that the sampling frequency was insufficient to properly assess ground contact during sprinting. Previous studies that have examined kinematics during high velocity running have used similar sensor sampling rates (Slaughter & Adamczyk, 2020; Vellucci & Beaudette, 2023). Velluci et al. used IMU sensors sampling at 60Hz to find coordinative strategies associated with over-ground sprinting (Vellucci & Beaudette, 2023), while Slaughter et al. used IMU sensors sampling at 60–100Hz to examine the knee kinematics of cutting motions during ultimate frisbee games. Both studies used sampling frequencies lower than the 128Hz used in this study to bolster the knowledge base of high speed running stance phase kinematics in the real world. We therefore believe that the results obtained in this study are minimally affected by the sampling rate of the IMUs.

5. Conclusion

This study gave further insight into how previous hamstring injury could affect the pelvis kinematics and stability during the stance phase of sprinting, demonstrating that mediolateral motion and asymmetry of the pelvis during stance of previously injured athletes was larger than that of healthy counterparts. Since the probability of sustaining a hamstring injury increases after a previous injury has been sustained, the alterations in ground contact dynamics on the previously injured leg may be a contributing factor to future re-injury.

Data availability

Data is available upon request.

Ethical approval

Participants provided written informed consent before participating in the study, and assent was obtained from the coaching staff at NCAA institutions. IRB approval was obtained from Northeastern University.

Funding

No outside funding was received for this study.

Declaration of competing interest

None declared.

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